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CLOCK & CALENDAR – COMPLETE SHORT NOTES

1. CLOCK BASICS

- A clock has **3 hands**:
 - **Hour hand** → short hand
 - **Minute hand** → long hand
 - **Second hand** → thin hand
- Clock face = **12 equal hour spaces** and **60 equal minute spaces**.
- **60 min = 1 hour**
- **30 min = $\frac{1}{2}$ hour**
- **15 min = $\frac{1}{4}$ hour**
- **1 min = 60 sec**

Important Angles

- 60 minute spaces = **360°**
- **1 minute space = 6°**
- 12 hour spaces = **360°**
- **1 hour space = 30°**
- Minute hand:
 - **6° per minute**
 - **360° per hour**
- Hour hand:
 - **$\frac{1}{2}^\circ$ per minute**
 - **30° per hour**

Clock Angle Formula

For time **h : m**:

$$\text{Angle} = |30h - 5.5m|$$

Equivalent:

$$\theta = |(11/2)m - 30h|$$

If the answer is **more than 180°** , use:

$$\text{Smaller angle} = 360^\circ - \text{obtained angle}$$

2. IMPORTANT CLOCK FACTS & TRICKS

Relative movement

- In **60 min**, minute hand gains **55 spaces** over the hour hand.
- Therefore, relative speed of minute hand over hour hand:

$5\frac{1}{2}^\circ$ per minute

- Minute hand gains **55 spaces in 60 min**.
- To gain **1 space**:

$$60/55 = 12/11 \text{ min}$$

Coincidence

The hands coincide **once in every hour**, but not exactly once during every clock hour.

- In **12 hours** → **11 coincidences**
- In **24 hours** → **22 coincidences**

Opposite directions

Hands are opposite when the angle between them is **180°** .

- They are opposite **11 times in 12 hours**
- **22 times in 24 hours**

Right angle

When hands are at **90°** , they are **15 minute-spaces apart**.

Straight line

When hands are in a straight line:

- Same direction → **0°**
 - Opposite direction → **180°**
-

3. CLOCK TYPE QUESTIONS

Type 1: Minute hand covers a distance

If minute hand covers **m minutes**:

$$\text{Angle} = m \times 6^\circ$$

Example:

80 minute-spaces:

$$80 \times 6^\circ = 480^\circ$$

Trick

Always convert the given distance into **minute spaces** first.

Type 2: Angle between hands at a given time

Use:

$$\theta = |30h - 5.5m|$$

Example: 8:35

Hour hand angle:

$$30 \times 8 + 35 \times \frac{1}{2} = 257.5^\circ$$

Minute hand angle:

$$35 \times 6 = 210^\circ$$

Difference:

$$47.5^\circ$$

Important

Do NOT assume the hour hand stays exactly on the number.

It moves continuously with the minutes.

Type 3: Hands inclined at a given time

Example: 15 min past 5

- Minute hand at 3
- Hour hand is slightly ahead of 5
- Distance between 3 and 5 = **2 spaces = 60°**
- Hour hand moves in 15 min:

$$15 \times \frac{1}{2} = 7.5^\circ$$

Therefore:

$$\text{Angle} = 60^\circ + 7.5^\circ = 67.5^\circ = 67\frac{1}{2}^\circ$$

Type 4: Clock gains/loses time

Basic idea

If a clock gains **10 min in 24 hours**:

$$\text{Clock's 24 h} = \text{Correct 24 h} + 10 \text{ min}$$

For a given clock reading, compare the **clock time** and **correct time** using a proportion.

Illustration

Clock is correct at **8 AM** and gains **10 min in 24 h**.

From 8 AM to next 1 PM = **29 hours**.

Clock gains:

$$10/24 \times 29 = 145/12 \text{ min}$$

$$= 12 \text{ min } 5 \text{ sec}$$

So when the clock shows **1 PM**, actual correct time is approximately:

12:48 PM

Shortcut

For a clock that gains/loses time:

$$\text{Clock time} : \text{Correct time} = \text{Clock's total elapsed time} : \text{Correct total elapsed time}$$

Always calculate the gain/loss over the **elapsed time**.

4. HANDS IN A STRAIGHT LINE BUT NOT TOGETHER

If hands are in a straight line but not together:

Required separation = 30 minute-spaces

Example: Between 7 and 8

At 7:00:

- Hour hand = 7
- Minute hand = 12
- Difference = **25 spaces**

To become opposite:

$$30 - 25 = 5 \text{ spaces}$$

Minute hand gains **55 spaces in 60 min**.

Therefore time required:

$$60/55 \times 5 = 60/11 = 5 \frac{5}{11} \text{ min}$$

Answer:

5 5/11 min past 7

Key Trick

For these questions:

1. Find initial separation.
2. Required separation for opposite hands = **30 spaces**.
3. Find extra spaces required.
4. Relative gain = **55 spaces/hour**.
5. Time = **Required spaces** $\times$ **60/55**.

- 2004 $\rightarrow$ Leap
- 2008 $\rightarrow$ Leap
- 2012 $\rightarrow$ Leap
- 1900 $\rightarrow$ **Not leap**
- 2000 $\rightarrow$ **Leap**
- 2100 $\rightarrow$ **Not leap**
- 2400 $\rightarrow$ **Leap**

Easy Rule

Divisible by 4 $\rightarrow$ Leap, except century years; century years must be divisible by 400.

CALENDAR

5. CALENDAR BASICS

- **1 day** = smallest unit
- **1 week** = **7 days**
- **1 month** = **12 part of a year**
- **1 year** = Earth's complete revolution around the Sun
- **100 years** = **1 century**

Ordinary Year

- **365 days**
- **365 = 52 weeks + 1 day**
- Therefore, ordinary year gives **1 odd day**.

Leap Year

- **366 days**
- **366 = 52 weeks + 2 days**
- Therefore, leap year gives **2 odd days**.

6. LEAP YEAR RULE — VERY IMPORTANT

A year is a leap year if:

1. It is divisible by 4.

BUT

2. Century years must be divisible by **400**.

Examples

7. ODD DAYS

Meaning

Odd days = days left after removing complete weeks.

Since:

7 days = 0 odd days

Important conversions

1 Ordinary year = 365 days = 52 weeks + 1 day $\rightarrow$ 1 odd day

1 Leap year = 366 days = 52 weeks + 2 days $\rightarrow$ 2 odd days

8. ODD DAYS IN CENTURIES

100 years

100 years = 76 ordinary + 24 leap years

Odd days:

$$76 \times 1 + 24 \times 2 = 124$$

$$124 \div 7 \rightarrow \text{remainder } 5$$

Therefore:

100 years = 5 odd days

200 years

$$5 \times 2 = 10 \rightarrow 3 \text{ odd days}$$

Therefore:

200 years = 3 odd days

300 years

$5 \times 3 = 15 \rightarrow 1$ odd day

Therefore:

300 years = 1 odd day

400 years

$5 \times 4 + 1 = 21 \rightarrow 0$ odd days

Therefore:

400 years = 0 odd days

MUST REMEMBER

Period	Odd Days
---------------	-----------------

100 years	5
-----------	---

200 years	3
-----------	---

300 years	1
-----------	---

400 years	0
-----------	---

400-year cycle = 0 odd days.

9. DAY ↔ ODD-DAY TABLE

Odd Days	Day
-----------------	------------

0	Sunday
---	--------

1	Monday
---	--------

2	Tuesday
---	---------

3	Wednesday
---	-----------

4	Thursday
---	----------

5	Friday
---	--------

6	Saturday
---	----------

Shortcut

Take total odd days **mod 7**, then use the table.

10. FINDING THE DAY OF A PARTICULAR DATE

Standard Method

Step 1

Choose a **reference day/date** whose day is known.

Step 2

Find the **total number of odd days** between the reference date and required date.

Step 3

If the required date is **after** the reference date:

Required day = Reference day + odd days

If required date is **before** the reference date:

Required day = Reference day – odd days

Important

When counting the number of days, **exclude the reference day**.

11. QUICK METHOD FOR A DATE

To find the day of a date:

Step 1 — Years

Count complete years before the required year and calculate their odd days.

- Ordinary year $\rightarrow 1$
- Leap year $\rightarrow 2$

Step 2 — Months

Add the number of days in the completed months of the required year.

Step 3 — Date

Add the required date.

Step 4

Convert total into odd days:

Total $\div 7 \rightarrow$ remainder

Step 5

Use the **Odd Days $\rightarrow$ Day table**.

12. MONTH DAYS TO REMEMBER

Ordinary Year

Month	Days
-------	------

January	31
---------	----

February	28
----------	----

March	31
-------	----

April	30
-------	----

May	31
-----	----

June	30
------	----

July	31
------	----

August	31
--------	----

September	30
-----------	----

October	31
---------	----

November	30
----------	----

December	31
----------	----

Leap Year

February = 29 days

Month Odd Days

- 31 days → **3 odd days**
- 30 days → **2 odd days**
- 29 days → **1 odd day**
- 28 days → **0 odd days**

13. SPECIAL CALENDAR FACTS

First day of a century

The first day of a century **cannot be:**

Wednesday, Friday or Sunday

Last day of a century

The last day of a century **cannot be:**

Tuesday, Thursday or Saturday

14. DAY AFTER TOMORROW / BEFORE YESTERDAY

Remember:

Today → Tomorrow → Day after tomorrow

Example:

If the day after tomorrow is **Sunday**:

- Tomorrow = Saturday
- Today = Friday
- Yesterday = Thursday
- Day before yesterday = **Wednesday**

Shortcut

Day before yesterday = 2 days before today.

15. CALENDAR REPEAT / SAME CALENDAR

Two years have the same calendar when:

1. Their dates fall on the same weekdays.
2. The total odd days between them = **0 mod 7**.
3. Be careful with **leap years**, because they contribute **2 odd days**.

Example: 1997

- 1997 → ordinary → 1 odd day
- 1998 → 1
- 1999 → 1
- 2000 → leap → 2
- 2001 → 1
- 2002 → 1

Total:

1 + 1 + 1 + 2 + 1 + 1 = 7 = 0 odd days

Therefore, **2003 has the same calendar as 1997.**

Key Trick

For same-calendar questions, calculate the odd days between the two years.

If remainder = 0, the calendars repeat.

16. EXAM-IMPORTANT CLOCK SHORTCUTS

Remember these numbers:

- 1 hour = 30°
- 1 minute = 6°
- Hour hand = $\frac{1}{2}^\circ/\text{min}$
- Minute hand = $6^\circ/\text{min}$
- Relative speed = $5\frac{1}{2}^\circ/\text{min}$
- Relative gain = 55 spaces/hour
- Coincidence = 11 times/12 hours
- Opposite = 11 times/12 hours
- Right angle = 90° = 15 spaces
- Straight line = 180° = 30 spaces
- Angle formula = $|30h - 5.5m|$

Day finding:

Years odd days + completed months odd days + date $\rightarrow$ divide by 7 $\rightarrow$ use day table.

Clock angle:

$$\theta = |30h - 5.5m|$$

Clock problems:

Always account for the **movement of the hour hand**; it does not remain fixed at the number.

17. EXAM-IMPORTANT CALENDAR SHORTCUTS

Remember:

- Ordinary year $\rightarrow$ 1 odd day
- Leap year $\rightarrow$ 2 odd days
- 100 years $\rightarrow$ 5 odd days
- 200 years $\rightarrow$ 3 odd days
- 300 years $\rightarrow$ 1 odd day
- 400 years $\rightarrow$ 0 odd days
- 7 odd days = 0
- 31-day month $\rightarrow$ 3 odd days
- 30-day month $\rightarrow$ 2 odd days
- 29-day February $\rightarrow$ 1 odd day
- 28-day February $\rightarrow$ 0 odd days

Golden Rules

Leap year: divisible by 4, except century years must be divisible by 400.

CLOCK & CALENDAR — PRACTICE QUESTIONS 1–16

1. How many times in a day are the hands of a clock pointing opposite to each other?

- (a) 20
- (b) 4
- (c) 24
- (d) 22

Solution:

The hands of a clock point in opposite directions **11 times in 12 hours**.

Therefore, in 24 hours:

$$11 \times 2 = 22$$

Answer: (d) 22

2. How many times from 4 pm to 10 pm, the hands of a clock are at right angle?

- (a) 11
- (b) 6
- (c) 90
- (d) 10

Solution:

Between 4 pm and 10 pm, the hands of a clock are at right angles **11 times**.

Answer: (a) 11

3. Find the angle traced by hour hand of a correct clock between 8 O'clock and 2 O'clock.

- (a) 180°
- (b) 150°
- (c) 210°
- (d) 200°

Solution:

Time period between 8 o'clock and 2 o'clock:

$$= 6 \text{ hours}$$

The hour hand moves 30° in 1 hour.

Therefore:

$$6 \times 30^\circ = 180^\circ$$

Answer: (a) 180°

4. What will be the angle between hour hand and minute hand, if clock shows 8:30 pm?

- (a) 90°
- (b) 75°
- (c) 60°
- (d) 85°

Solution:

Formula:

$$\text{Angle} = |(11/2 \times \text{minutes}) - (30 \times \text{hours})|$$

Here:

$$\text{Hours} = 8$$

$$\text{Minutes} = 30$$

$$\text{Angle} = |(11/2 \times 30) - (30 \times 8)|$$

$$= |165 - 240|$$

$$= 75^\circ$$

Answer: (b) 75°

5. At half past 5 in the evening, the smaller angle between the hour and minute hands of a clock is

- (a) 10°
- (b) 12°
- (c) 15°
- (d) 18°

Solution:

Formula:

$$\text{Angle} = |(11/2 \times \text{minutes}) - (30 \times \text{hours})|$$

Here:

$$\text{Hours} = 5$$

$$\text{Minutes} = 30$$

$$\text{Angle} = |(11/2 \times 30) - (30 \times 5)|$$

$$= |165 - 150|$$

$$= 15^\circ$$

Answer: (c) 15°

6. The angle between the hour hand and the minute hand of a clock when time is 3:25 will be

- (a) $47\frac{1}{2}^\circ$
- (b) $42\frac{1}{2}^\circ$
- (c) 60°
- (d) 45°

Solution:

Formula:

$$\text{Angle} = |(11/2 \times \text{minutes}) - (30 \times \text{hours})|$$

Here:

Hours = 3

Minutes = 25

$$\text{Angle} = |(11/2 \times 25) - (30 \times 3)|$$

$$= |137.5 - 90|$$

$$= 47.5^\circ$$

$$= 47\frac{1}{2}^\circ$$

Answer: (a) $47\frac{1}{2}^\circ$

7. A clock was set correct at 12 o'clock. It loses 10 minutes per hour. What will be the angle between the hour and minute hand of the clock after one hour?

- (a) 90°
- (b) 85°
- (c) 75°
- (d) 105°

Solution:

The clock loses 10 minutes per hour.

Therefore, after 1 actual hour, the clock will show:

$$60 - 10 = 50 \text{ minutes}$$

So, the clock shows **12:50**.

$$\text{Angle} = |(11/2 \times 50) - (30 \times 12)|$$

$$= |275 - 360|$$

$$= 85^\circ$$

Answer: (b) 85°

8. At what time, in minutes, between 3 O'clock and 4 O'clock will the hands of the clock coincide with each other?

- (a) $5 \frac{1}{11}$
- (b) $12 \frac{4}{11}$

- (c) $14 \frac{3}{11}$
- (d) $16 \frac{4}{11}$

Solution:

At 3 o'clock, the minute hand is **15 spaces** away from the hour hand.

To coincide, the minute hand must gain 15 spaces.

The minute hand gains **55 spaces in 60 minutes**.

Therefore:

$$\text{Time} = 60/55 \times 15$$

$$= 12/11 \times 15$$

$$= 180/11$$

$$= 16 \frac{4}{11} \text{ minutes}$$

Therefore, the hands coincide at:

16 $\frac{4}{11}$ minutes past 3

Answer: (d) $16 \frac{4}{11}$

9. At what time between 5 and 6, are the hands of a clock at coincident?

- (a) 22 min past 5
- (b) 30 min past 5
- (c) $27 \frac{3}{11}$ min past 5
- (d) $31 \frac{3}{11}$ min past 5

Solution:

At 5 o'clock, the minute hand is **25 spaces** away from the hour hand.

To coincide, the minute hand must gain 25 spaces.

The minute hand gains **55 spaces in 60 minutes**.

Therefore:

$$\text{Time} = 60/55 \times 25$$

$$= 12/11 \times 25$$

$$= 300/11$$

$$= 27 \frac{3}{11} \text{ minutes}$$

Therefore, the hands coincide at:

27 $\frac{3}{11}$ minutes past 5

Answer: (c) $27 \frac{3}{11}$

10. At what time between 4 and 5 O'clock, will the hands of a watch point in opposite directions?

- (a) 45 min past 4
- (b) 40 min past 4
- (c) 50 $\frac{4}{11}$ min past 4
- (d) 54 $\frac{6}{11}$ min past 4

Solution:

At 4 o'clock, the hands are **20 spaces** apart.

For opposite directions, they must be **30 spaces** apart.

Therefore, the minute hand must gain:

$$20 + 30 = 50 \text{ spaces}$$

The minute hand gains 55 spaces in 60 minutes.

Therefore:

$$\text{Time} = \frac{60}{55} \times 50$$

$$= \frac{600}{11}$$

$$= 54 \frac{6}{11} \text{ minutes}$$

Therefore, the hands will be opposite at:

54 $\frac{6}{11}$ minutes past 4

Answer: (d) 54 $\frac{6}{11}$

11. At what time between 5:30 and 6 O'clock, will the hands of a clock be at right angle?

- (a) 43 $\frac{3}{11}$ min past 5
- (b) 43 $\frac{7}{11}$ min past 5
- (c) 40 min past 5
- (d) 45 min past 5

Solution:

At 5 o'clock, the hands are **25 spaces** apart.

For a right angle, the hands must be **15 spaces** apart.

Therefore, the minute hand needs to gain:

$$25 + 15 = 40 \text{ spaces}$$

The minute hand gains 55 spaces in 60 minutes.

Therefore:

$$\text{Time} = \frac{60}{55} \times 40$$

$$= \frac{480}{11}$$

$$= 43 \frac{7}{11} \text{ minutes}$$

Therefore, the hands will be at right angle at:

43 $\frac{7}{11}$ minutes past 5

Answer: (b) 43 $\frac{7}{11}$

12. Two clocks are set correctly at 9 am on Monday. Both the clocks gain 3 min and 5 min respectively in an hour. What time will the second clock register, if the first clock which gains 3 min in an hour shows the time as 6:30 pm on the same day?

- (a) 6:27 pm
- (b) 6:45 pm
- (c) 6:25 pm
- (d) 6:50 pm
- (e) None of these

Solution:

From 9 am to 6:30 pm:

$$= 9 \text{ hours } 30 \text{ minutes}$$

$$= 570 \text{ minutes}$$

The first clock gains 3 minutes per hour.

Therefore, it runs at:

$$60 + 3 = 63 \text{ minutes per actual hour}$$

Actual time elapsed:

$$\frac{570}{63} \text{ hours}$$

The second clock gains 5 minutes per hour.

Therefore, it runs at:

$$60 + 5 = 65 \text{ minutes per actual hour}$$

For the same actual elapsed time, the second clock shows:

$$\frac{65}{63} \times 570 \text{ minutes}$$

$$= 587.5 \text{ minutes}$$

$$= 9 \text{ hours } 47.5 \text{ minutes}$$

Therefore, the second clock shows approximately:

6:45 pm

Answer: (b) 6:45 pm

13. A clock is set right at 5 am. The clock loses 16 min in 24 h. What will be the right time when the clock indicates 10 pm on the 4th day?

- (a) 11:15 pm
- (b) 11:00 pm

- (c) 12:00 pm
(d) 12:30 pm

Solution:

From 5 am to 10 pm on the 4th day:

= 89 hours

The clock loses 16 minutes in 24 hours.

Therefore, 24 hours of correct time corresponds to:

24 hours – 16 minutes

= 23 hours 44 minutes of the faulty clock.

Thus:

23 hours 44 minutes of faulty clock = 24 hours of correct clock.

Therefore, 89 hours of faulty clock corresponds to:

$$(24 \times 60 \times 89) / (23 \times 60 + 44)$$

= 90 hours of correct time

Therefore:

5 am + 90 hours = 11 pm

Answer: (b) 11:00 pm

14. The minute hand of a clock overtakes the hour hand at intervals of 63 min on the correct time. How much does a clock gain or lose in a day?

- (a) $54 \frac{8}{77}$ min (gain)
(b) $56 \frac{8}{77}$ min (gain)
(c) $53 \frac{8}{77}$ min (gain)
(d) $53 \frac{8}{77}$ min (gain)

Solution:

In a correct clock, the minute hand gains **55 spaces in 60 minutes**.

To gain 60 spaces:

$$\text{Time} = 60/55 \times 60$$

= $65 \frac{5}{11}$ minutes

The given clock takes 63 minutes.

Therefore, gain in 63 minutes:

$$65 \frac{5}{11} - 63$$

= $2 \frac{5}{11}$ minutes

= $27/11$ minutes

Gain in 24 hours:

$$= 27/11 \times (60 \times 24)/63$$

$$= 4320/77$$

= $56 \frac{8}{77}$ minutes

Since the result is positive, the clock **gains**.

Answer: (b) $56 \frac{8}{77}$ min (gain)

CALENDAR

15. The day before the day before yesterday is three days after Saturday. What day is it today?

- (a) Tuesday
(b) Wednesday
(c) Thursday
(d) Friday

Solution:

Three days after Saturday is:

Saturday → Sunday → Monday → Tuesday

Therefore:

Day before the day before yesterday = Tuesday.

So:

Day before yesterday = Wednesday

Yesterday = Thursday

Today = Friday

Answer: (d) Friday

16. The last day of a century cannot be either:

- I. Tuesday
II. Thursday
III. Saturday
IV. Sunday

(a) I and II
(b) I and IV
(c) I, II and III
(d) I, III and IV

Solution:

The last day of a century cannot be:

- Tuesday
- Thursday

- Saturday

Therefore:

I, II and III

Answer: (c) I, II and III

CLOCK & CALENDAR — PRACTICE QUESTIONS 17–33

CLOCK & CALENDAR — PRACTICE QUESTIONS 17–33

17. If today is Thursday, then what will be the day on 363rd day?

- (a) Sunday
- (b) Saturday
- (c) Thursday
- (d) None of these

Solution:

$$\begin{aligned} &[\\ 363 &= 51 \times \text{weeks} + 6 \times \text{days} \\ &] \end{aligned}$$

Therefore:

$$\text{Thursday} + 6 \text{ days} = \text{Wednesday}$$

Wednesday is not given in the options.

Answer: (d) None of these

18. Today is Monday. After 61 days, it will be

- (a) Wednesday
- (b) Saturday
- (c) Tuesday
- (d) Thursday

Solution:

Since every 7 days make one complete week:

$$61 = 8 \text{ weeks} + 5 \text{ days}$$

Therefore:

$$\text{Monday} + 5 \text{ days} = \text{Saturday}$$

Answer: (b) Saturday

19. If it is Saturday on January 1, 2000, then January 1, 2001 would have been

- (a) Monday
- (b) Tuesday
- (c) Friday
- (d) Saturday
- (e) None of these

Solution:

2000 is a **leap year**.

A leap year has:

$$366 \text{ days} = 52 \text{ weeks} + 2 \text{ days}$$

Therefore:

$$\text{Saturday} + 2 \text{ days} = \text{Monday}$$

Answer: (a) Monday

20. If January 1 is a Friday, then what is the first day of the month of March in a leap year?

- (a) Tuesday
- (b) Wednesday
- (c) Thursday
- (d) Friday

Solution:

In a leap year:

$$\text{January} = 31 \text{ days}$$

$$\text{February} = 29 \text{ days}$$

Total:

$$31 + 29 = 60 \text{ days}$$

$$60 = 8 \text{ weeks} + 4 \text{ days}$$

Therefore:

$$\text{Friday} + 4 \text{ days} = \text{Tuesday}$$

Hence, March 1 is Tuesday.

Answer: (a) Tuesday

21. In a month of 31 days, third Thursday falls on 16th. What will be the last day of the month?

- (a) 5th Friday
- (b) 4th Saturday
- (c) 5th Wednesday
- (d) 5th Thursday

Solution:

Given:

$$16\text{th} = \text{Thursday}$$

The last day is 31st.

Difference:

$$31 - 16 = 15 \text{ days}$$

$$15 = 2 \text{ weeks} + 1 \text{ day}$$

Therefore:

Thursday + 1 day = Friday

So, 31st is Friday.

Since it is a 31-day month, this is the **5th Friday**.

Answer: (a) 5th Friday

22. If Republic Day was celebrated in 1996 on Friday, on which day in 2000 Independence was celebrated?

- (a) Tuesday
- (b) Monday
- (c) Sunday
- (d) Saturday

Solution:

Republic Day = 26 January.

Given:

26 January 1996 = Friday

The total odd days between 26 January 1996 and 15 August 2000 give a shift of **4 days**.

Therefore:

Friday + 4 days = Tuesday

Hence, Independence Day in 2000 was Tuesday.

Answer: (a) Tuesday

23. If 18th February, 1997 falls on Tuesday, then what will be the day on 18th February, 1999?

- (a) Monday
- (b) Tuesday
- (c) Thursday
- (d) Friday

Solution:

1997 is an ordinary year:

= 1 odd day

1998 is also an ordinary year:

= 1 odd day

Total:

$1 + 1 = 2$ odd days

Given:

18 February 1997 = Tuesday

Therefore:

Tuesday + 2 days = Thursday

Answer: (c) Thursday

24. If, in year 2000, birthday of Netaji be Wednesday, what was the Independence Day in that year?

- (a) Wednesday
- (b) Thursday
- (c) Friday
- (d) Saturday

Solution:

Netaji's birthday = 23 January.

Given:

23 January 2000 = Wednesday

Independence Day = 15 August.

Number of days from 23 January to 15 August:

$8 + 29 + 31 + 30 + 31 + 30 + 31 + 15$

= 205 days

$205 = 29 \text{ weeks} + 2 \text{ days}$

Therefore:

Wednesday + 2 days = Friday

Answer: (c) Friday

25. If 27 March, 1995 was a Monday, then what day of the week was 1 November, 1994?

- (a) Sunday
- (b) Monday
- (c) Tuesday
- (d) Wednesday

Solution:

Number of days from 1 November 1994 to 27 March 1995:

November 1994 = 30 days

December 1994 = 31 days

January 1995 = 31 days

February 1995 = 28 days

March 1995 = 27 days

Total:

$$30 + 31 + 31 + 28 + 27 = 147 \text{ days}$$

$$147 = 21 \text{ weeks} + 0 \text{ days}$$

Therefore, both dates fall on the same day.

Given:

27 March 1995 = Monday

Hence:

1 November 1994 = Monday

Answer: (b) Monday

26. What was the day on 2nd July, 1984?

- (a) Wednesday
- (b) Tuesday
- (c) Monday
- (d) Thursday

Solution:

Odd days in 1600 years = 0

Odd days in 300 years = 1

Odd days in 1900 years = 1

For 83 years:

40 leap years + 43 ordinary years

Odd days:

$$40 \times 2 + 43 \times 1$$

$$= 123 \text{ odd days}$$

$$123 = 17 \text{ weeks} + 4 \text{ days}$$

Therefore, total odd days up to 1983:

$$1 + 4 = 5$$

From 1 January 1984 to 2 July 1984:

$$31 + 29 + 31 + 30 + 31 + 30 + 2$$

$$= 184 \text{ days}$$

$$184 = 26 \text{ weeks} + 2 \text{ days}$$

Total odd days:

$$5 + 2 = 7$$

$$= 0 \text{ odd days}$$

Therefore, the day is the same as the reference day.

Answer: (c) Monday

27. It is observed that January 1, 2023 is a Sunday. In which year the January 1st will fall on a Sunday?

- (a) 2029
- (b) 2030
- (c) 2034
- (d) 2037

Solution:

Odd days from 2023 to 2033:

$$2023 = 1$$

$$2024 = 2$$

$$2025 = 1$$

$$2026 = 1$$

$$2027 = 1$$

$$2028 = 2$$

$$2029 = 1$$

$$2030 = 1$$

$$2031 = 1$$

$$2032 = 2$$

$$2033 = 1$$

Total:

$$1 + 2 + 1 + 1 + 1 + 2 + 1 + 1 + 1 + 2 + 1$$

$$= 14 \text{ days}$$

$$14 = 2 \text{ weeks} = 0 \text{ odd days}$$

Therefore, January 1, 2034 will fall on the same day as January 1, 2023.

Answer: (c) 2034

28. The year next to 1990 will have the same calendar as that of the year 1990.

- (a) 1995
- (b) 1997
- (c) 1996
- (d) 1992

Solution:

1990 = ordinary year = 1 odd day

1991 = 1 odd day

1992 = 2 odd days

1993 = 1 odd day

1994 = 1 odd day

1995 = 1 odd day

Total:

$$1 + 1 + 2 + 1 + 1 + 1 = 7 \text{ days}$$

$$7 \text{ days} = 0 \text{ odd days}$$

Therefore, 1996 will have the same calendar as 1990.

Answer: (c) 1996

29. I was born on August 11. Mohan is older than me by 11 days. This year's Independence Day falls on Monday. The day on which Mohan's birthday will fall this year will be

- (a) Monday
- (b) Tuesday
- (c) Sunday
- (d) Thursday

Solution:

Mohan is 11 days older.

Therefore, Mohan's birthday is:

$$11 \text{ days before August 11} = \text{July 31}$$

Given:

$$15 \text{ August} = \text{Monday}$$

$$\text{August 1 is 14 days before August 15.}$$

Since 14 is exactly 2 weeks:

$$\text{August 1} = \text{Monday}$$

Therefore:

$$\text{July 31} = \text{Sunday}$$

Answer: (c) Sunday

30. Mrs. Susheela celebrated her wedding anniversary on Tuesday, 30th September, 1997. When will she celebrate her next wedding anniversary on the same day?

- (a) 30th September, 2003
- (b) 30th September, 2004
- (c) 30th September, 2002
- (d) 30th October, 2003

Solution:

Calculate the odd days from 1997 to 2003:

$$1997-98 = 1$$

$$1998-99 = 1$$

$$1999-2000 = 2$$

$$2000-01 = 1$$

$$2001-02 = 1$$

$$2002-03 = 1$$

Total:

$$1 + 1 + 2 + 1 + 1 + 1 = 7$$

$$7 \text{ days} = 0 \text{ odd days.}$$

Therefore, the same day will occur on:

30 September 2003

Answer: (a) 30th September, 2003

31. If the Republic Day of India in 1980 falls on Saturday, X was born on March 3, 1980. Y is 4 days older to X, then Y's birthday fell on

- (a) Thursday
- (b) Friday
- (c) Wednesday
- (d) None of these

Solution:

Republic Day:

$$26 \text{ January } 1980 = \text{Saturday}$$

Number of days from 26 January to 3 March:

$$5 + 29 + 3 = 37 \text{ days}$$

$$37 = 5 \text{ weeks} + 2 \text{ days}$$

Therefore:

$$3 \text{ March } 1980 = \text{Saturday} + 2 \text{ days}$$

$$= \text{Monday}$$

Y is 4 days older than X.

Therefore, Y's birthday was 4 days before Monday:

$$\text{Monday} - 4 \text{ days} = \text{Thursday}$$

Answer: (a) Thursday

32. Nasir's birthday is on Thursday May 18. On what day of the week will be Rehan's birthday in the same year if Rehan was born on August 19?

- (a) Wednesday
- (b) Saturday
- (c) Friday
- (d) Thursday

Solution:

Given:

18 May = Thursday

Number of days from 18 May to 19 August:

$$13 + 30 + 31 + 19$$

$$= 93 \text{ days}$$

$$93 = 13 \text{ weeks} + 2 \text{ days}$$

Therefore:

Thursday + 2 days = Saturday

Hence, August 19 was Saturday.

Answer: (b) Saturday

33. The birthday of Y was celebrated six days before X, who was born on October 4, 1999. The Independence day of that year fell on Sunday. On which day did Y celebrate her birthday, if it was not a leap year?

- (a) Wednesday
- (b) Tuesday
- (c) Monday
- (d) Sunday

Solution:

X was born on:

4 October 1999

Y's birthday was 6 days before X.

Therefore:

$$4 \text{ October} - 6 \text{ days} = 28 \text{ September}$$

Given:

15 August 1999 = Sunday

Number of days from 15 August to 28 September:

$$16 + 28 = 44 \text{ days}$$

$$44 = 6 \text{ weeks} + 2 \text{ days}$$

Therefore:

Sunday + 2 days = Tuesday

Hence, Y celebrated her birthday on Tuesday.

Answer: (b) Tuesday